modenanalyse_2fe2s

Version 1.0.0

Manual and Theoretical Documentation

Mode-resolved bonding reorganization analysis
for [2Fe-2S] iron-sulfur proteins
from Gaussian-16 and ORCA-6 frequency calculations

Lukas Knauer
AG Schünemann
RPTU Kaiserslautern-Landau

Cluster type:	[2Fe-2S], reduced or oxidized, for Rieske, ferredoxin, and mitoNEET ligation
Input:	Gaussian-16 `.log` (`freq=hpmodes` preferred; standard block as fallback) or ORCA-6 `.hess` PDB file with same cluster (optional, recommended)
Output:	Excel reports (mode analysis, SCSD, reorganization energies, modulation spectra, UMAP clusters), REPORT (plain text), plots (PNG)
External packages:	`numpy`, `scipy`, `pandas`, `matplotlib`, `openpyxl`, `scikit-learn`, `umap-learn`, `hdbscan`, `scsdpy`
References:	Marcus, *J. Chem. Phys.* **24**, 966 (1956) Marcus, *Angew. Chem. Int. Ed. Engl.* **32**, 1111 (1993) Hammes-Schiffer & Soudackov, *J. Phys. Chem. B* **112**, 14108 (2008) Huang & Rhys, *Proc. R. Soc. Lond. A* **204**, 406 (1950) Kingsbury & Senge, *Chem. Sci.* **15**, 13638 (2024)
License:	GNU General Public License v3.0 or later (`GPL-3.0-or-later`, see `LICENSE`)
Copyright:	© 2026 Lukas Knauer, AG Schünemann, RPTU Kaiserslautern-Landau

May 7, 2026

Contents

1 Introduction

`modenanalyse_2fe2s` is a Python package for mode-resolved bonding reorganization analysis of [2Fe-2S] iron-sulfur clusters from DFT frequency calculations. For each vibrational mode and each relevant bonding channel, it computes thermally scaled bond modulations Δr_X and the associated Marcus-Hush reorganization contributions λ_X in a form directly comparable to spectroscopic observables (NRVS, NIS).

1.1 What the tool does

The pipeline takes a Gaussian-16 `.log` or ORCA-6 `.hess` frequency calculation of a [2Fe-2S]-containing protein region (and optionally a PDB file for ligand identification) and produces:

- **Per-mode analysis**: Eigenvector-based classification of all normal modes into in-plane / out-of-plane / torsional based on the cluster plane.
- **Reorganization energies** per bonding channel $X \in \{\text{Fe-Fe}, \text{Fe-S}, \text{Fe-N}, \text{N-H}, \text{H-acceptor}\}$ and per mode, both as system totals and per-bond resolved.
- **Frequency-resolved modulation spectra** $M_X(\omega)$ suitable for direct comparison with NRVS / NIS measurements.
- **Cumulative reorganization** $\Lambda_X(\omega)$ for identifying which spectral regions dominate.
- **SCSD symmetry-coordinate decomposition** of the cluster core into D_{2h} irreps (after Kingsbury & Senge, 2024).
- **UMAP embeddings** of the 31-feature mode space with HDBSCAN clustering for mode-landscape exploration.
- **Multi-cluster support**: dimers and multi-[2Fe-2S] systems are handled as a first-class feature.

1.2 Target users

The intended users are researchers in biophysical chemistry, bioinorganic chemistry, and quantum chemistry who study iron-sulfur proteins. Typical applications include:

- NRVS band assignment in Rieske-type proteins, mitoNEET, and ferredoxins.
- Comparative reorganization-energy studies between mutants or oxidation states.
- Identification of vibrational modes relevant to PCET.
- Validation of QM/MM models against vibrational data.

1.3 Scope and limitations

The tool assumes a single [2Fe-2S] cluster (or multiple, with `analyze_all_clusters=True`) per input system. It does not currently handle [4Fe-4S], [3Fe-4S], or larger iron-sulfur clusters — these would require different cluster-detection logic and different SCSD reference geometries.

The reorganization formalism is local-mode based: each individual bond is treated independently via the Marcus-Hush expression $\lambda = \frac{1}{2}\mu\omega^2(\Delta r)^2$. Couplings between bonding channels are not currently included beyond what is implicit in the DFT normal modes themselves.

2 Theoretical Background

2.1 The Marcus-Hush reorganization energy

In Marcus electron-transfer theory, the reorganization energy λ measures the energy cost of distorting the equilibrium geometry of one electronic state to match the equilibrium geometry of another. For inner-sphere reorganization in a single bond at frequency ω with reduced mass μ and bond-length change Δr, the harmonic approximation gives

$$\lambda = \tfrac{1}{2}\,\mu\,\omega^2\,(\Delta r)^2. \tag{1}$$

For a multi-mode system, the total reorganization is the sum of contributions from individual normal modes:

$$\Lambda_X = \sum_i \lambda_X(i), \tag{2}$$

where the sum runs over normal modes i and X is a bonding channel (Fe-Fe, Fe-N, etc.). This is the central observable produced by this tool, written to the `Reorg_total` sheet.

2.2 Thermal scaling of eigenvectors

DFT normal modes are normalized; to convert into physical displacements, we use the quantum-mechanical RMS amplitude of a harmonic oscillator at temperature T:

$$u_{\mathrm{rms}}^2 = \frac{\hbar}{2m\omega}\coth\left(\frac{\hbar\omega}{2k_BT}\right), \tag{3}$$

where m is the mode reduced mass and $\omega = 2\pi c\tilde{\nu}$ in SI units. At $T \to 0$, coth $\to 1$ and $u_{\mathrm{rms}}^2 \to \hbar/(2m\omega)$ (the zero-point amplitude). At high temperature, coth $\to 1/x$ and $u_{\mathrm{rms}}^2 \to k_BT/(m\omega^2)$ (classical equipartition).

For NRVS measurements, $T = 5$ K is the standard low-temperature condition; $T = 80$ K is also common. Room temperature (300 K) is relevant only for room-temperature kinetics studies.

2.3 Bond-channel modulation Δr_X

For each mode i and each bond $X = (a, b)$, the bond-length modulation is computed from the thermally scaled atomic displacements $\boldsymbol{e}_a, \boldsymbol{e}_b$:

$$\Delta r_X(i) = (\boldsymbol{e}_a - \boldsymbol{e}_b) \cdot \hat{r}_{ab}, \tag{4}$$

where $\hat{r}_{ab}$ is the unit vector along the equilibrium bond. This is a signed scalar (positive = bond stretching, negative = bond compression).

For multiple sub-channels (e.g., FeN with two His ligands), the per-channel reorganization sums over sub-channels:

$$\lambda_X(i) = \tfrac{1}{2}\,\mu\,\omega_i^2 \sum_k w_k\,[\Delta r_{X,k}(i)]^2, \tag{5}$$

with weights w_k accounting for symmetry-equivalent sub-channels (e.g., two equivalent His ligands each contributing $w = 1/2$).

2.4 Hydrogen-bond reaction coordinate (PCET)

For the H-acceptor (HA) channel relevant to PCET, the relevant coordinate is not a single bond length but the projection of the hydrogen displacement relative to the donor onto the donor-acceptor axis:

$$\Delta r_{\mathrm{HA}}(i) = (\boldsymbol{e}_H - \boldsymbol{e}_{\mathrm{donor}}) \cdot \hat{r}_{\text{donor-acceptor}}. \tag{6}$$

This captures the actual proton-transfer reaction coordinate. A nonzero Λ_{HA} indicates PCET capability of the system.

2.5 Asymptotic limit at $T \to 0$

In the zero-temperature limit, eq. (3) gives $u_{\mathrm{rms}} = \sqrt{\hbar/(2m\omega)}$, and eq. (1) becomes

$$\lambda_X(T \to 0) = \frac{\hbar\omega}{4} \cdot \alpha_X^2, \tag{7}$$

where $\alpha_X = \Delta r_X / u_{\mathrm{rms}}$ is the dimensionless mode-bond coupling. For a single atom moving along the bond axis with amplitude $\alpha = 1$, this simplifies to $\lambda = \omega/4$ (in $\hbar\omega$ units, so $\omega/4$ in cm^{-1} when frequency is in cm^{-1}). This is the Huang-Rhys form of vibronic coupling and is the asymptotic check the implementation passes.

2.6 Choice of reduced mass

Two conventions are produced:

- **Pair-reduced mass** $\mu_{\text{pair}} = m_a m_b/(m_a + m_b)$: treats each bond as an isolated diatomic. Useful for direct interpretation of bond-vibrational energies.
- **Mode-reduced mass** μ_{mode}: the full DFT normal-mode reduced mass, which incorporates couplings to other atoms. Produced as `Lambda_total_mode_cm1` in the output. *This is the recommended quantity for Marcus-Hush comparisons across systems*, as it correctly includes the inertia of the actual collective motion.

The two conventions agree when the mode is a pure local bond vibration; they differ when the mode is delocalized across multiple bonds.

2.7 D_{2h} symmetry-coordinate decomposition (SCSD)

The cluster core (4 atoms: 2 Fe + 2 S) has approximate D_{2h} symmetry: the 12 atomic Cartesian degrees of freedom decompose as

$$\Gamma_{\text{core}} = A_g \oplus B_{1g} \oplus B_{2g} \oplus B_{3g} \oplus A_u \oplus B_{1u} \oplus B_{2u} \oplus B_{3u} \oplus 4\,T_{\text{trans+rot}}. \tag{8}$$

The SCSD method (Kingsbury & Senge, 2024) projects the actual displacement of the cluster core onto a canonical D_{2h} reference geometry (Fe-Fe = 2.73 Å, Fe-S = 2.20 Å, taken as means from high-resolution Rieske/ferredoxin crystal structures), with axis convention x = Fe-Fe direction, y = S-S direction, z = cluster normal. The decomposition is orthogonal in D_{2h} irreps; values are directly comparable between different structures and oxidation states.

3 Installation and First Run

3.1 Installation

The package ships as a Python wheel:

```
pip install modenanalyse_2fe2s-1.0.0-py3-none-any.whl
```

Dependencies are pulled in automatically (numpy, scipy, pandas, matplotlib, openpyxl, scikit-learn, umap-learn, hdbscan, scsdpy).

Verify installation:

```
modenanalyse-2fe2s --help
```

3.2 Quick test

The package ships with a small test [2Fe-2S] cluster:

```
from modenanalyse_2fe2s import Config, run_analysis

cfg = Config(
    log_file = "tests/data/Cys_2Fe-2S_red3_hpfrq_opt.log.xz",
    output_dir = "./quickstart_results",
    temp_k = 5.0,
    freq_max = 800.0,
)
run_analysis(cfg)
```

This takes about 1 minute and produces the full output set.

3.3 Recommended TOML-based workflow

For reproducibility, prefer TOML configuration files:

```
modenanalyse-2fe2s --write-template my_run.toml
# edit my_run.toml as needed
modenanalyse-2fe2s --config my_run.toml
```

The TOML file documents itself and is git-friendly.

4 Configuration Reference

The `Config` dataclass has 48 fields. Most have sensible defaults; only `log_file` and `output_dir` are required.

4.1 Required fields

- `log_file`: path to Gaussian-16 `.log` (with `freq=hpmodes`) or ORCA-6 `.hess`.
- `output_dir`: target directory; created if absent.

4.2 Geometry input

- `pdb_file` (optional): for ligand-residue identification via Kabsch-aligned matching to the QM cluster.
- `pdb_chain`: chain identifier (default `"A"`; empty string `""` = all chains).

4.3 Frequency window

- `freq_min`, `freq_max`: frequency filter in cm^{-1}. Defaults: `None` (no limit) and 800.0 (NRVS measurement window).
- `freq_windows`: list of `(low, high)` tuples for multi-window analysis. Each window gets its own subfolder; a top-level overall analysis is also produced.

4.4 Temperature

- `temp_k`: sample temperature in Kelvin. Standard values: 5 K (cryogenic NRVS), 40 K (helium standard), 80 K (liquid nitrogen), 300 K (room temperature). Default: 80.

4.5 Cluster selection

- `cluster_index`: 0-based index of the cluster to analyze (default 0 = closest Fe-Fe pair). Ignored when `analyze_all_clusters=True`.
- `analyze_all_clusters`: `True` to analyze all detected [2Fe-2S] clusters, each in its own `cluster_N/` subfolder, with a top-level `multi_cluster_summary.txt`.

4.6 Module selection

- `analyze_scsd`: enable SCSD D_{2h} decomposition (default `True`).
- `pcet_enabled`: enable PCET reorganization pipeline (default `True`).

4.7 Numerical parameters

- `sigma_eigvec`: noise amplitude added to eigenvectors for sensitivity analysis (default 5×10^{-4}).
- `sigma_coord`: noise amplitude added to atomic coordinates (default 1×10^{-3}).
- `interp_step`: grid spacing for interpolated spectra in cm^{-1} (default 0.05).
- `interp_boundary_mode`: `"context"`, `"zero"`, or `"nearest"` (default `"context"`).
- `amplitude`: fallback amplitude in classical mode (`temp_k=None`); default 1.0.

4.8 PCET geometry

- `pcet_acceptor_r0_a`: H-bond acceptor reference distance (default 2.8 Å).
- `pcet_acceptor_sigma_a`: weighting sigma (default 0.4 Å).
- `pcet_hbond_cutoff_a`: maximum H-acceptor distance for inclusion (default 3.5 Å).

4.9 Output controls

- `include_hydrogen`: include H atoms in mode analysis (default `True`).
- `use_cache`: speed up repeated runs on the same log file via byte-offset cache (default `True`).

5 Output Files Reference

5.1 REPORT.txt

Human-readable summary, written as `<basename>_REPORT.txt`. Sections (in order): CONFIGURATION, INSTALLED MODULES, GEOMETRY, COORDINATING AMINO ACIDS, MODE ANALYSIS, MODE DISTRIBUTION, CLUSTER ANALYSIS, FAILED MODES, WARNINGS, ERRORS, NOTES, OUTPUT FILES.

The NOTES section is the densest with information — it includes the headline reorganization values, the SCSD methodology citation, HP/standard consistency check, and analysis statistics.

5.2 Main analysis workbook

Written as `<basename>_analysis.xlsx`, with 11 sheets:

Sheet	Content
`Mode_analysis`	Per-mode summary (one row per mode)
`Core_scores`	Cluster-core mode scores
`Equilibrium_distances`	Fe-Fe, Fe-S, Fe-N, etc. distances
`SCSD`	D_{2h} symmetry decomposition
`Reorganization_energy`	Per-mode $\lambda_X(i)$ and $\Delta r_X(i)$
`Reorg_total`	System totals Λ_X
`Reorg_per_bond`	Per-bond contributions
`Modulation_spectra`	$M_X(\omega)$
`Lambda_cumulative`	$\Lambda_X(\omega)$
`B_factors`	Debye-Waller B-factors
`Info`	Run metadata

5.2.1 Mode_analysis

One row per analyzed mode. Columns:

- `Mode#`, `Frequency` (cm^{-1}), `u_rms` (Å), `Red.mass` (AMU), `Frc.const.` (mDyn/Å): from DFT.

- `Sym.`: irrep label.
- `Type`: classification (`In-plane`, `Out-of-plane`, `Torsional/Mixed`).
- `Type (fine)`: 7-level classification (`Pure OOP`, `Strong OOP`, `Majority OOP`, `Mixed`, `Majority INP`, `Strong INP`, `Pure INP`).
- `Prec.`: precision (`high` for HP eigenvector blocks, `standard` otherwise).
- Ring-2 / Ring-3 / Ring-1 columns: percentages of mode amplitude on ligand atoms / secondary sphere / cluster core.
- `Lambda_FeFe`, `Lambda_NH`, `Lambda_HA` (cm^{-1}): per-mode reorganization contributions for the three most informative channels.

5.2.2 Reorg_total — the headline numbers

Five rows, one per channel:

Channel	**Λ_pair (cm^{-1})**	**Λ_mode (cm^{-1})**	**n_modes**
FeFe	...	...	...
FeN	...	...	...
FeS	...	...	...
NH	...	...	...
HA	...	...	...

Use `Lambda_total_mode_cm1` for cross-system comparisons. A non-zero Λ_{HA} indicates PCET capability.

5.2.3 Reorganization_energy — per-mode breakdown

One row per mode, one column block per channel. For each channel X:

- `dr_X_rss_a`: root-sum-of-squares of bond modulations across sub-channels (signed sum is also available as `dr_X_sum_signed_a`).
- `lambda_X_pair_cm1`: λ with pair-reduced mass.
- `lambda_X_mode_cm1`: λ with mode-reduced mass.

5.2.4 Modulation_spectra

Frequency-resolved $M_X(\omega) = \sum_i \delta(\omega - \omega_i) \cdot |\Delta r_X(i)|^2$ broadened with a Gaussian. Useful for direct visual comparison with NRVS spectra.

5.2.5 Lambda_cumulative

$\Lambda_X(\omega) = \sum_{i:\omega_i \leq \omega} \lambda_X(i)$. Plotted versus ω, this shows which spectral regions contribute most to the total reorganization.

5.3 Embeddings workbook

`<basename>_analysis_Embeddings.xlsx`: PCA, UMAP, and t-SNE projections of the 31-feature mode space, with HDBSCAN cluster labels and per-cluster characterization (Z-scores, Fisher F).

5.4 Plots

`<basename>_embedding_UMAP.png`: 2D projection of all modes, left panel colored by frequency, right panel colored by mode type.

6 Multi-Cluster Workflow

For dimers (e.g., glutaredoxin) and multi-domain systems with multiple [2Fe-2S] clusters, set `analyze_all_clusters=True`:

```
cfg = Config(
    log_file = "dimer.hess",
    pdb_file = "dimer.pdb",
    output_dir = "./results_dimer",
    analyze_all_clusters = True,
    temp_k = 5.0,
)
run_analysis(cfg)
```

The pipeline:

1. Detects all [2Fe-2S] clusters in the input.
2. For each cluster, runs the full analysis in a separate `cluster_N/` subfolder, with cluster-specific Fe-Fe and Fe-S identification.
3. Writes a top-level `multi_cluster_summary.txt` with geometry and run status per cluster.

Multi-cluster Kabsch alignment

For multi-cluster systems, the Kabsch alignment of QM coordinates to PDB requires care: with >2 Fe atoms in the PDB, the algorithm chooses the Fe pair whose Fe-Fe distance is closest to the QM Fe-Fe distance *and* is below 4 Å (cluster-internal distance). This avoids accidental cross-cluster alignment that would otherwise produce silently wrong ligand assignments.

Single-cluster fallback

For single-cluster systems, `analyze_all_clusters=True` falls back automatically to single-cluster mode (no subfolders).

7 Validation

The implementation has been validated against:

7.1 Numerical formula verification

All core formulas were independently re-implemented and checked to six decimal places:

- Eq. (3) (thermal amplitude): zero-point limit matches at $T \to 0$, classical limit converges at $T \to \infty$.
- Eq. (1) (Marcus-Hush): $T \to 0$ with $\alpha = 1$ gives $\lambda = \omega/4$ exactly.
- Eq. (7) (low-T asymptotic): consistency check verified at multiple frequencies and reduced masses.
- Kabsch SVD: standard reflection-corrected variant, verified against a known reference.

7.2 Test system: mitoNEET-H87C

Validated against a QM/MM Hessian of mitoNEET-H87C (Cys_4 ligation, 422 atoms, 1266 modes after filter, 11 UMAP clusters):

Quantity	Value
Λ_{FeFe} (mode)	≈ 29 cm^{-1}
Λ_{FeS} (mode)	≈ 338 cm^{-1}
Λ_{FeN} (mode)	0
Λ_{NH} (mode)	0
Λ_{HA} (mode)	0
Cluster Fe-Fe distance	2.78 Å
Fe-S (mean)	2.26 Å
Fe-S-Fe (mean)	75.9°
Folding residual	0.053 Å

The H87C mutation removes the only His ligand, so Λ_{NH} and Λ_{HA} are both zero — as expected. The near-equality $\Lambda_{\mathrm{FeN}} = 0$ is a consistency check.

7.3 Edge cases tested

- Empty atom list, all-Fe (no S), single Fe pair, multi-cluster dimer (4 Fe + 4 S), 3 Fe with no S — all handled correctly.
- Degenerate cluster geometries (collinear atoms): protected against division by zero with safety checks.
- Multi-cluster Kabsch with 4 PDB Fe atoms: verified to choose the correct intra-cluster pair.

7.4 Test suite

The package includes 92+ tests:

- 88+ unit tests covering geometry, reorganization formulas, Kabsch alignment, classification logic, multi-cluster wrapper.
- 4 end-to-end smoke tests that run a complete analysis on a test [2Fe-2S] cluster and verify the output structure and key numerical values.

Run with:

```
pytest tests/ # ~5 sec
pytest tests/ -m slow # ~70 sec, full pipeline
```

8 Troubleshooting

"No [2Fe-2S] cluster found"

Check that the input contains atomic coordinates. For Gaussian `.log`, the file must have a Standard orientation block. For ORCA `.hess`, the `$atoms` block must be present and non-empty.

If the cluster has unusually large Fe-Fe (>3.5 Å) or Fe-S (>2.7 Å) distances, increase the cutoffs:

```
cfg = Config(..., fe_fe_cutoff=4.0, fe_s_cutoff=3.0)
```

"Kabsch alignment failed"

Common causes:

- Wrong chain identifier. Default is `"A"`; if your PDB uses a different chain, set `pdb_chain="X"` or `pdb_chain=""` (all chains).

- PDB has fewer than 2 Fe or 2 S atoms.
- PDB and QM coordinates are in completely different reference frames and the Fe-Fe distance differs by more than the algorithm can tolerate.

The REPORT shows the Kabsch RMSD; values < 0.1 Å indicate a good alignment.

"Λ_{NH} and Λ_{HA} are zero on a Rieske protein"

Check:

1. PDB has the His ligands present.
2. PDB is correctly aligned to the QM region (Kabsch RMSD in REPORT).
3. `pcet_enabled=True` (the default).
4. The His residues' protonation: the tool checks for H on the ring nitrogen.

If His ligands are present but `Lambda_HA` is zero, the WARNINGS section will indicate that no H-bond acceptors were found within the cutoff distance — typically meaning no nearby protein groups can act as proton acceptor.

"openpyxl missing"

`pip install openpyxl`. Required for Excel output. Pinned in the package dependencies but may need refresh after Python upgrades.

Slow runs on large systems

For systems with >1500 modes, the per-mode analysis is the bottleneck. The bottleneck scales linearly with N_{modes}. A typical 422-atom system with 1266 modes runs in ≈5 minutes on modern hardware. Use `use_cache=True` (default) to speed up repeated runs on the same input.

9 Citation and License

9.1 License

`modenanalyse_2fe2s` is licensed under the GNU General Public License v3.0 or later (GPL-3.0-or-later). See `LICENSE` for the full text.

The dependencies have their own licenses; users who redistribute the package should consult the dependency manifest for license compatibility.

9.2 Citation

If you use `modenanalyse_2fe2s` in a publication, please cite as in `CITATION.cff`. After Zenodo upload, the package will have a permanent DOI:

```
Knauer, L. (2026). modenanalyse_2fe2s: Mode-resolved bonding
reorganization analysis for [2Fe-2S] iron-sulfur clusters from DFT
frequency calculations (Version 1.0.0) [Software].
DOI: <to be assigned upon first GitHub release>
```

9.3 Scientific references

The implementation rests on the following methodological literature:

- Marcus, R. A. *J. Chem. Phys.* **24**, 966 (1956) — foundation of electron-transfer theory.
- Marcus, R. A. *Angew. Chem. Int. Ed. Engl.* **32**, 1111 (1993) — Nobel-lecture review.
- Hammes-Schiffer, S. & Soudackov, A. V. *J. Phys. Chem. B* **112**, 14108 (2008) — PCET formalism.
- Huang, K. & Rhys, A. *Proc. R. Soc. Lond. A* **204**, 406 (1950) — vibronic coupling foundations.
- Kingsbury, C. J. & Senge, M. O. *Chem. Sci.* **15**, 13638 (2024) — SCSD methodology.
- Gee, L. B., Pelmenschikov, V., Mons, C. *et al.* *Biochemistry* **60**, 2419 (2021) — mitoNEET NRVS.